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Scientific Interests

Circumstellar disks, planetary systems, radio interferometry, scientific software development

Education

The Pennsylvania State University, University Park, PA <i>Ph.D., Astronomy & Astrophysics</i>	2020-2023
The Pennsylvania State University, University Park, PA <i>M.S., Astronomy & Astrophysics</i>	2018-2020 GPA: 3.83
Lycoming College, Williamsport, PA <i>B.S., Physics (Minors: Astronomy, Mathematics)</i>	2014-2018 GPA: 4.0

Research Appointments

Constraining vertical structures of debris disks in ARKS <i>Advisor: Dr. Meredith Hughes</i>	Wesleyan University 2023-present
Regularized maximum likelihood imaging for ALMA <i>Advisor: Dr. Ian Czekala</i>	The Pennsylvania State University 2020-2023
Migration traps as the root cause of the Kepler dichotomy <i>Advisors: Dr. Eric Ford, Dr. Daniel Carrera</i>	The Pennsylvania State University 2021-2022
Rapid formation of super-Earths around low-mass stars <i>Advisors: Dr. Eric Ford, Dr. Daniel Carrera</i>	The Pennsylvania State University 2018-2021
Detecting nonlinearity in binary star data <i>Advisor: Dr. Christopher Kulp</i>	Lycoming College 2018
Using missing ordinal patterns to detect nonlinearity in time series data <i>Advisor: Dr. Christopher Kulp</i>	Lycoming College 2017-2018
The connection between solar coronal cavities and solar filaments <i>Advisors: Dr. Kathy Reeves, Dr. Nishu Karna, and Jakub Prchlik</i>	Harvard-Smithsonian CfA 2017

First-Author Publications

- [5] *The ALMA survey to Resolve exoKuiper belt Substructures (ARKS) III: The Vertical Structure of Debris Disks*, **Brianna Zawadzki** et al. 2026, A&A, 705, A197.
- [4] *exoALMA. IX. Regularized Maximum Likelihood Imaging of Non-Keplerian Features*, **Brianna Zawadzki** et al. 2025, ApJL, 984, L14.
- [3] *Regularized Maximum Likelihood Image Synthesis and Validation for ALMA Continuum Observations of Protoplanetary Disks*, **Brianna Zawadzki**, et al. 2023, PASP, 135 064503.
- [2] *Migration traps as the root cause of the Kepler dichotomy*, **Brianna Zawadzki**, Daniel Carrera, and Eric Ford 2022, ApJ, 937, 53.
- [1] *Rapid Formation of Super-Earths Around Low-Mass Stars*, **Brianna Zawadzki**, Daniel Carrera, and Eric Ford 2021, MNRAS, 503, 1.

Contributing-Author Publications

- [34] *Viscously Stirring Particle Disks into Lorentzians and Gaussians to Infer Dynamical and Collisional Masses (ARKS XIII)*, Eugene Chiang et al. incl. **Brianna Zawadzki** 2026, ApJ, 1008, 163.
- [33] *The ALMA survey to Resolve exoKuiper belt Substructures (ARKS): XI. Gas-dust interactions and radial offsets between micron and millimetre-sized grains*, Johan Olofsson et al. incl. **Brianna Zawadzki** 2026, A&A, 710, A265.
- [32] *exoALMA. XXIII. Estimating Disk and Planet Properties from Dust Morphologies with DBNets 2.0*, Alessandro Ruzza et al. incl. **Brianna Zawadzki** 2026, ApJL 1000 L16.
- [31] *exoALMA XXII: A Two-dimensional Atlas of Deviations from Keplerian Disks*, Misato Fukagawa et al. incl. **Brianna Zawadzki** 2026, ApJL 1000 L15.
- [30] *exoALMA. XXI. The Morphology and Dynamics of Vertical Flows*, Myriam Benisty et al. incl. **Brianna Zawadzki** 2026, ApJL 1000 L14.
- [29] *exoALMA. XX. Tomographic Detection of Embedded Planets in Protoplanetary Disks*, Andrés F. Izquierdo, et al. incl. **Brianna Zawadzki** 2026, ApJL 1000 L13.
- [28] *exoALMA. XIX. Confirmation of Non-thermal Line Broadening in the DM Tau Protoplanetary Disk*, Caitlyn Hardiman et al. incl. **Brianna Zawadzki** 2026, ApJL, 997, L47.
- [27] *The ALMA survey to Resolve exoKuiper belt Substructures (ARKS). X: Interpreting the peculiar dust rings around HD 131835*, Marija R. Jankovic et al. incl. **Brianna Zawadzki** 2026, A&A, 705, A204.
- [26] *The ALMA survey to Resolve exoKuiper belt Substructures (ARKS). IX: Gas-driven origin for the continuum arc in the debris disc of HD 121617*, Philipp Weber et al. incl. **Brianna Zawadzki** 2026, A&A, 705, A203.
- [25] *The ALMA survey to Resolve exoKuiper belt Substructures (ARKS). VIII: A dust arc and non-Keplerian gas kinematics in HD 121617*, Sebastian Marino et al. incl. **Brianna Zawadzki** 2026, A&A, 705, A202.
- [24] *The ALMA survey to Resolve exoKuiper belt Substructures (ARKS). VII: Optically Thick Gas with Broad CO Gaussian Local Line Profiles in the HD 121617 Disc*, Aoife Brennan et al. incl. **Brianna Zawadzki** 2026, A&A, 705, A201.
- [23] *The ALMA survey to Resolve exoKuiper belt Substructures (ARKS). VI: Asymmetries and Offsets*, Joshua Lovell et al. incl. **Brianna Zawadzki** 2026, A&A, 705, A200.
- [22] *The ALMA survey to Resolve exoKuiper belt Substructures (ARKS). V: Comparison between scattered light and thermal emission*, Julien Milli et al. incl. **Brianna Zawadzki** 2026, A&A, 705, A199.
- [21] *The ALMA survey to Resolve exoKuiper belt Substructures (ARKS). IV: CO Gas Imaging and Overview*, Sorcha MacManamon et al. incl. **Brianna Zawadzki** 2026, A&A, 705, A198.
- [20] *The ALMA survey to Resolve exoKuiper belt Substructures (ARKS). II: The Radial Structure of Debris Discs*, Yinuo Han et al. incl. **Brianna Zawadzki** 2026, A&A, 705, A196.
- [19] *The ALMA survey to Resolve exoKuiper belt Substructures (ARKS). I: Motivation, Sample, Data Reduction and Results Overview*, Sebastian Marino et al. incl. **Brianna Zawadzki** 2026, A&A, 705, A195.
- [18] *exoALMA. XVIII. Interpreting large scale kinematic structures as moderate warping*, Andrew Winter et al. incl. **Brianna Zawadzki** 2025, ApJL, 990, L10.
- [17] *exoALMA. XVII. Characterizing the Gas Dynamics around Dust Asymmetries*, Lisa Wölfer et al. incl. **Brianna Zawadzki** 2025, ApJL, 984, L22.
- [16] *exoALMA. XVI. Predicting Signatures of Large-scale Turbulence in Protoplanetary Disks*, Marcelo Barraza-Alfaro et al. incl. **Brianna Zawadzki** 2025, ApJL, 984, L21.
- [15] *exoALMA. XV. Interpreting the Height of CO Emission Layer*, Giovanni P. Rosotti et al. incl. **Brianna Zawadzki** 2025, ApJL, 984, L20.
- [14] *exoALMA. XIV. Gas Surface Densities in the RX J1604.3–2130 A Disk from Pressure-broadened CO Line Wings*, Tomohiro C. Yoshida et al. incl. **Brianna Zawadzki** 2025, ApJL, 984, L19.
- [13] *exoALMA. XIII. Gas Masses from N_2H^+ and $C^{18}O$: A Comparison of Measurement Techniques for Protoplanetary Gas Disk Masses*, Leon Trapman et al. incl. **Brianna Zawadzki** 2025, ApJL, 984, L18.

- [12] *exoALMA. XII. Weighing and Sizing exoALMA Disks with Rotation Curve Modelling*, Cristiano Longarini et al. incl. **Brianna Zawadzki** 2025, ApJL, 984, L17.
- [11] *exoALMA. XI. ALMA Observations and Hydrodynamic Models of LkCa 15: Implications for Planetary Mass Companions in the Dust Continuum Cavity*, Charles H. Gardner et al. incl. **Brianna Zawadzki** 2025, ApJL, 984, L16.
- [10] *exoALMA. X. Channel Maps Reveal Complex 12CO Abundance Distributions and a Variety of Kinematic Structures with Evidence for Embedded Planets*, Christophe Pinte et al. incl. **Brianna Zawadzki** 2025, ApJL, 984, L15.
- [9] *exoALMA. VIII. Probabilistic Moment Maps and Data Products Using Nonparametric Linear Models*, Thomas Hilder et al. incl. **Brianna Zawadzki** 2025, ApJL, 984, L13.
- [8] *exoALMA. VI. Rotating under Pressure: Rotation Curves, Azimuthal Velocity Substructures, and Gas Pressure Variations*, Jochen Stadler et al. incl. **Brianna Zawadzki** 2025, ApJL, 984, L11.
- [7] *exoALMA. V. Gaseous Emission Surfaces and Temperature Structures*, Maria Galloway-Sprietsma et al. incl. **Brianna Zawadzki** 2025, ApJL, 984, L10.
- [6] *exoALMA. IV. Substructures, Asymmetries, and the Faint Outer Disk in Continuum Emission*, Pietro Curone et al. incl. **Brianna Zawadzki** 2025, ApJL, 984, L9.
- [5] *exoALMA. III. Line-intensity Modeling and System Property Extraction from Protoplanetary Disks*, Andrés F. Izquierdo et al. incl. **Brianna Zawadzki** 2025, ApJL, 984, L8.
- [4] *exoALMA. II. Data Calibration and Imaging Pipeline*, Ryan A. Loomis et al. incl. **Brianna Zawadzki** 2025, ApJL, 984, L7.
- [3] *exoALMA. I. Science Goals, Project Design, and Data Products*, Richard Teague et al. incl. **Brianna Zawadzki** 2025, ApJL, 984, L6.
- [2] *An extreme test case for planet formation: a close-in Neptune orbiting an ultracool star*, Guðmundur Stefánsson et al. including **Brianna Zawadzki** 2023, Science, 382, 6674, pp. 1031-1035.
- [1] *Using missing ordinal patterns to detect nonlinearity in time series data*, Christopher W. Kulp, Luciano Zunino, Thomas Osborne, and **Brianna Zawadzki** 2017, Physical Review E 96, 022218.

Presentations

July 31, 2026 Talk	Discs on the Exe, Exeter, England: <i>Probing the Collisional Cascade in AU Mic with Resolved Vertical Structures at Multiple Wavelengths</i>
July 14, 2026 Talk	Debris Disk Connections, Cambridge, England: <i>Probing the Collisional Cascade in AU Mic with Resolved Vertical Structures at Multiple ALMA Wavelengths</i>
May 15, 2026 Talk	New York Area Exoplanets Meeting, New York, NY: <i>Probing the Collisional Cascade in AU Mic with Resolved Vertical Structures at Multiple Wavelengths</i>
Apr 28, 2026 Talk	Center for Astrophysics Harvard & Smithsonian, Cambridge, MA: <i>Probing the Collisional Cascade in AU Mic with Resolved Vertical Structures at Multiple Wavelengths</i>
Jul 21, 2025 Talk	Northeast Star & Planet Formation Meeting, New Haven, CT: <i>Characterizing Vertical Dust Structures in Debris Disks with ARKS</i>
Jun 15-20, 2025 Poster	Gordon Research Conference, South Hadley, MA: <i>Characterizing Vertical Dust Structures in Debris Disks with ARKS</i>
Jun 14, 2025 Talk	Gordon Research Seminal, South Hadley, MA: <i>Characterizing Vertical Dust Structures in Debris Disks with ARKS</i>
May 5-8, 2025 Poster	Planets on the Edge, Santa Barbara, CA: <i>Characterizing Vertical Dust Structures in Debris Disks with ARKS</i>
Jan 30, 2025 Invited Poster	BASP Frontiers Conference, Villars-sur-Ollon, Switzerland: <i>Searching for Protoplanets with ALMA Using RML Imaging Techniques</i>
Jan 14, 2025 Talk	AAS 245, National Harbor, MD: <i>ARKS III: Vertical Dust Structures in Debris Disks</i>

July 24, 2024 Talk	Connecticut Exoplanet Picnic, Middletown, CT: <i>Resolving Vertical Structures in Millimeter Debris Disk Observations with ARKS</i>
May 30, 2024 Talk	New York Area Exoplanets Meeting, New York, NY: <i>Resolving Vertical Structures in Millimeter Debris Disk Observations with ARKS</i>
Apr 26, 2024 Invited Talk	Williams College, Williamstown, MA: <i>Resolving Vertical Structures in Millimeter Debris Disk Observations with ARKS</i>
Apr 19, 2024 Talk	Center for Astrophysics Harvard & Smithsonian, Cambridge, MA: <i>Resolving Vertical Structures in Millimeter Debris Disk Observations with ARKS</i>
Apr 16, 2024 Talk	Columbia University, New York, NY: <i>Resolving Vertical Structures in Millimeter Debris Disk Observations with ARKS</i>
Mar 25, 2024 Talk	Dust Devils: Debris Disks in the Sonoran Desert, Tucson, AZ: <i>Resolving Vertical Structures in Millimeter Debris Disk Observations with ARKS</i>
Mar 18, 2024 Poster	Extreme Solar Systems V, Christchurch, New Zealand: <i>A High-Resolution View of Planet Formation Signatures in exoALMA Protoplanetary Disks</i>
Mar 5, 2024 Talk	Monash Stars and Planets Seminar, Melbourne, Australia: <i>A High-Resolution View of Planet Formation Signatures in exoALMA Protoplanetary Disks</i>
Feb 6, 2024 Talk	Yale Astronomy Exoplanets & Stars Seminar, New Haven, CT: <i>A High-Resolution View of Planet Formation Signatures in exoALMA Protoplanetary Disks</i>
Jan 12, 2023 Talk, Virtual	VLTI and ALMA Synthesis Imaging Workshop, Garching, Germany: <i>RML Imaging Techniques for ALMA Protoplanetary Disk Observations</i>
Oct 12, 2022 Poster	Institute for Computational and Data Sciences Symposium, State College, PA: <i>Regularized Maximum Likelihood Techniques for ALMA</i>
May 31, 2022 Talk	APEx Exocoffee, Heidelberg, Germany: <i>Regularized Maximum Likelihood Techniques for ALMA</i>
May 3, 2022 Talk	Exoplanets IV Conference, Las Vegas, NV: <i>Migration Traps as the Root Cause of the Kepler Dichotomy</i>
May 2, 2022 Poster	Exoplanets IV Conference, Las Vegas, NV: <i>Regularized Maximum Likelihood Techniques for ALMA</i>
Feb 25, 2022 Talk, Virtual	Submillimeter Array (SMA) Science Seminar: <i>Regularized Maximum Likelihood Techniques for ALMA</i>
Oct 6, 2021 Talk	North American ALMA Science Center, Charlottesville, VA <i>Regularized Maximum Likelihood Techniques for ALMA</i>
May 26, 2021 Talk, Virtual	Emerging Researchers in Exoplanet Science Conference <i>Regularized Maximum Likelihood Techniques for ALMA Spectral Line Imaging</i>
Sep 28, 2020 Poster, Virtual	Europlanet Science Congress <i>Rapid Formation of Super-Earths Around Low-Mass Stars</i>
Jul 29, 2020 Poster, Virtual	Exoplanets III Conference <i>Rapid Formation of Super-Earths Around Low-Mass Stars</i>
Jul 29, 2019 Poster	TESS Science Conference, Cambridge, MA <i>Rapid Formation of Super-Earths Around Low-Mass Stars</i>
Feb 11, 2019 Talk	The Pennsylvania State University, State College, PA <i>Rapid Formation of Super-Earths Around Low-Mass Stars</i>
Dec 11, 2017 Poster	American Geophysical Union Fall Meeting, New Orleans, LA <i>The Connection Between Solar Coronal Cavities and Solar Filaments</i>
Aug 9, 2017 Talk	Center for Astrophysics Harvard & Smithsonian, Cambridge, MA <i>The Connection Between Solar Coronal Cavities and Solar Filaments</i>

Teaching and Work Experience

ASTRO 105: Exploring the Cosmos <i>Instructor</i>	Wesleyan University <i>Spring 2026</i>
ASTRO 420W: Planets and Planetary System Formation <i>Taught the online component of the course, graded writing assignments</i>	The Pennsylvania State University <i>Fall 2020</i>
Exoplanets and the Search for Life Beyond Earth <i>Instructor</i>	PSU Upward Bound Virtual Summer Academy <i>Summer 2020</i>
ASTRO 414: Stellar Structure and Evolution <i>Graded homework assignments</i>	The Pennsylvania State University <i>Spring 2020</i>
ASTRO 402W: Astronomical Telescopes, Techniques, and Data Analysis <i>Facilitated and evaluated student telescope use</i>	The Pennsylvania State University <i>Spring 2020</i>
ASTRO 475W: Stars and Galaxies <i>Facilitated in-class discussion, graded writing assignments</i>	The Pennsylvania State University <i>Fall 2019</i>
ASTR 112: Fundamentals of Geology <i>Laboratory Assistant</i>	Lycoming College <i>Spring 2018</i>
ASTR 111: Fundamentals of Astronomy <i>Laboratory Assistant</i>	Lycoming College <i>Fall 2017</i>
Planetarium Operator <i>Gave occasional public planetarium shows</i>	Lycoming College Detwiler Planetarium <i>Spring 2017 - Spring 2018</i>
Academic Resource Center Tutor <i>Provided walk-in tutoring services for most mathematics courses, with special hours for multivariable calculus and differential equations</i>	Lycoming College <i>Fall 2016 - Spring 2018</i>
Outgassing Services International <i>Intern, QCM thermogravimetric analysis testing and analysis of GC/MS data</i>	Mountain View, CA <i>Summer 2016</i>
PHYS 226: Fundamentals of Physics II <i>Laboratory Assistant</i>	Lycoming College <i>Spring 2016, Spring 2017</i>
PHYS 225: Fundamentals of Physics I <i>Laboratory Assistant</i>	Lycoming College <i>Fall 2015, Fall 2016</i>

Outreach, Leadership, and Involvement

Wesleyan Star Gay-zing <i>Co-organizer and volunteer</i>	Biannually 2023 - present <i>Middletown, CT</i>
Kids Nights at Van Vleck Observatory <i>Volunteer</i>	Fall 2023 - present <i>Middletown, CT</i>
Space Nights at Van Vleck Observatory <i>Speaker</i>	October 2024 <i>Middletown, CT</i>
Astronomy on Tap: State College <i>Co-leader</i>	January 2021 - April 2023 <i>State College, PA</i>
Women and Underrepresented Genders in Astronomy (W+IA) <i>Co-leader from Fall 2020 - Spring 2023</i>	Fall 2018 - Spring 2023 <i>The Pennsylvania State University</i>
Towards A More Inclusive Astronomy (TaMIA) <i>General member</i>	Fall 2018 - Spring 2023 <i>The Pennsylvania State University</i>
Society of Physics Students <i>President in 2017, Vice-President in 2016</i>	Fall 2014 - May 2018 <i>Lycoming College</i>
STEM Affinity Community <i>President</i>	April 2017 - May 2018 <i>Lycoming College</i>

Honors, Awards, and Fellowships

Brinson Prize Postdoctoral Fellowship <i>Awarded to early-career astrophysicists to support innovative research.</i>	2023-present Wesleyan University
Science Achievement Graduate Fellowship Nominee <i>For contributions to the advancement of women in sciences.</i>	2022 The Pennsylvania State University
Center For Exoplanets and Habitable Worlds Grant <i>Awarded to fund travel and participation at Exoplanets IV Conference.</i>	2022 The Pennsylvania State University
AAS International Travel Grant <i>Awarded to students presenting at international science meetings.</i>	2020 The American Astronomical Society
Center For Exoplanets and Habitable Worlds Grant <i>Awarded to fund travel and participation at TESS Science Conference.</i>	2019 The Pennsylvania State University
University Graduate Fellowship <i>Awarded by the Eberly College of Science before the first year of graduate study.</i>	2018-2019 The Pennsylvania State University
The Charles J. Kocian Award <i>Awarded to the graduating senior with the highest GPA in the class.</i>	May 2018 Lycoming College
The Edward J. Gray Prize <i>Awarded to the individuals with the highest or second highest GPA in the senior class.</i>	May 2018 Lycoming College
Φυσικά Award in Astronomy & Physics <i>Given to the graduating senior with the highest departmental GPA.</i>	May 2018 Lycoming College
Dean's List <i>Awarded for maintaining a GPA of at least 3.5.</i>	Fall 2014-2017; Spring 2015-2018 Lycoming College
Kappa Mu Epsilon <i>National math honor society</i>	Inducted March 2017 Lycoming College
Sigma Pi Sigma <i>National physics honor society</i>	Inducted March 2016 Lycoming College
M.B. Rich Endowed Prize <i>Awarded to first-year students who complete their first year with a 4.0 GPA.</i>	April 2015 Lycoming College
Fundamentals of Physics Award <i>Awarded to the student who earns the highest grades in the introductory physics sequence.</i>	April 2015 Lycoming College
Principles of Astronomy Award <i>Awarded to the student who earns the highest grade in introductory astronomy.</i>	April 2015 Lycoming College

Mentoring

Sayed Shafaat Mahmud (B.A. Colgate University, 2026) <i>Project: Applying RML imaging techniques to debris disk observations</i> <i>Outcomes: Poster honorable mention at Wesleyan Summer Science Symposium, currently a PhD student in Physics at Carnegie Mellon University</i>	Summer 2025
Millicent Barnstone (B.A. Wellesley College, expected 2027) <i>Project: Resolving the vertical structure of the two rings in the HD 15115 debris disk</i> <i>Outcomes: Paper in prep, to be submitted to AAS journals</i>	Summer 2026